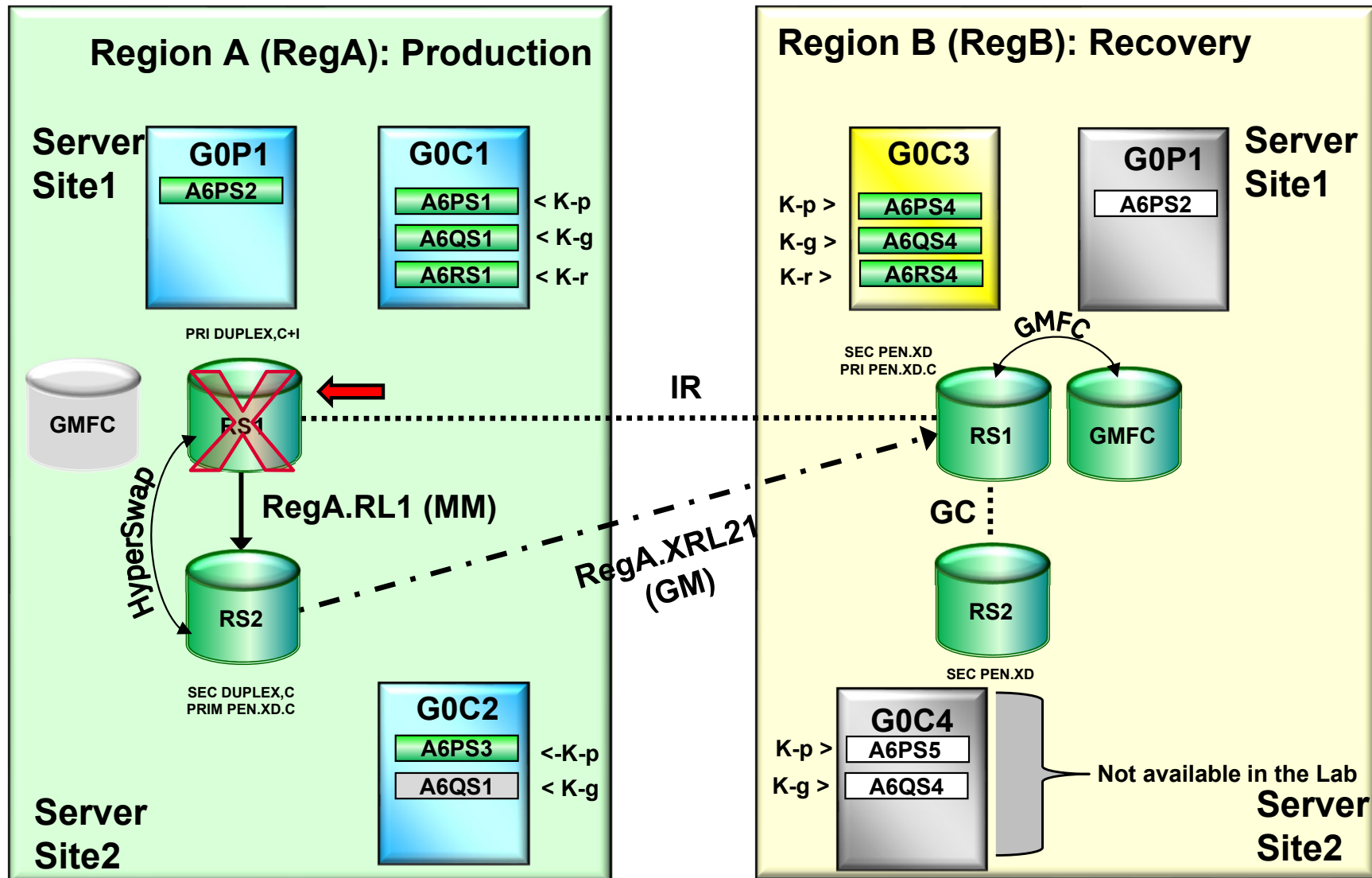


GDPS METRO GLOBAL – LAB: PRIMARY DISK OUTAGE



Unplanned primary disk outage (Cascade mode)

- triggering a HyperSwap with initial configuration MGM cascaded
 - Starting configuration



Unplanned primary disk outage

- triggering a HyperSwap with initial configuration MGM cascaded

▪ High level steps

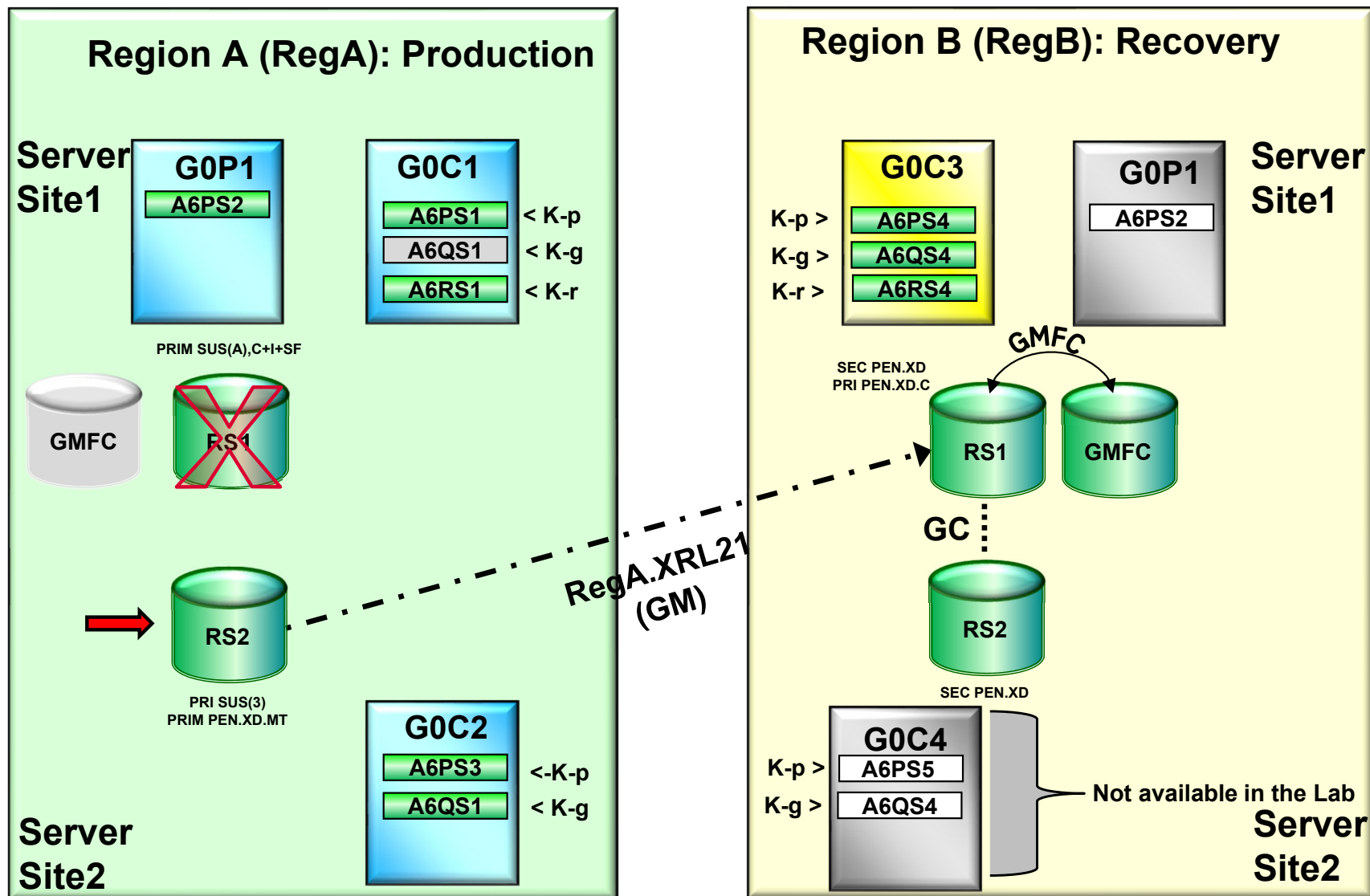
- Primary disk failure (SETIOS HYPERSWAP) triggered unplanned HyperSwap from RS1 to RS2 in region RegA
- Metro Mirror between RegA.RS1 and RegA.RS2 is frozen (FRZ)
- Application system GOP1 running on RegA.RS2
- Takeover script SWAPSITE1 executed
- Acknowledge GDPS notification of the event
- GM session not moved, running from RegA.RS2 to RegB.RS1
 - DR protection maintained

▪ Additional considerations to be covered after outage

- if SIMPLEX DASD in RS1 affected, move Kg-Sys to G0C2

Unplanned primary disk outage

- triggering a HyperSwap with initial configuration MGM cascaded
- Target configuration



Unplanned primary disk outage – return to normal

- to initial configuration MGM cascaded

▪ Option (1)

- Run control script to establish Metro Mirror RegA.RS2 > RegA.RS1 with START SECONDARY on Kp-Sys, entering a MT configuration
- Run control script with Planned HyperSwap RegA.RS2 > RegA.RS1 on Master Kp-Sys, coming back to a cascaded configuration with primary MM disk in RegA.RS1
- Run control script with START SECONDARY RegA.RS1 > RegA.RS2 to establish Metro Mirror



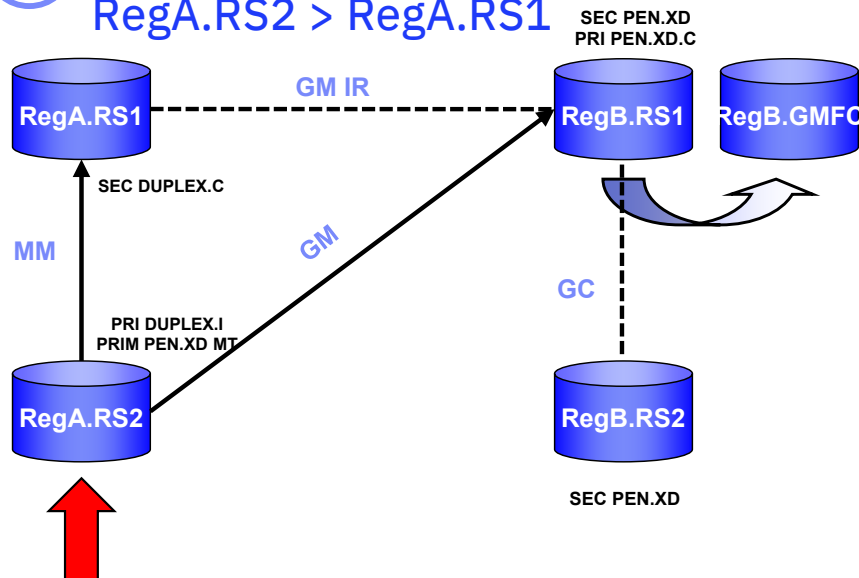
minimal management effort (shortest route)



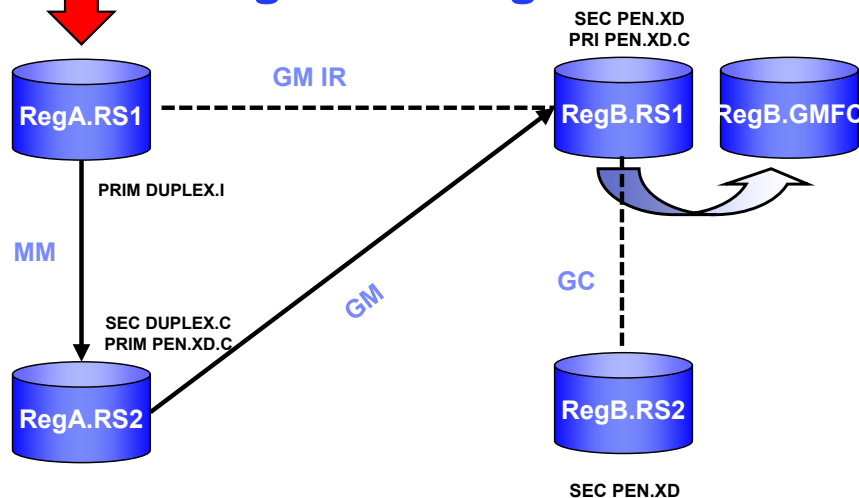
DR position ages from HyperSwap until START SECONDARY completes

Return to normal – MGM cascaded - Option (1)

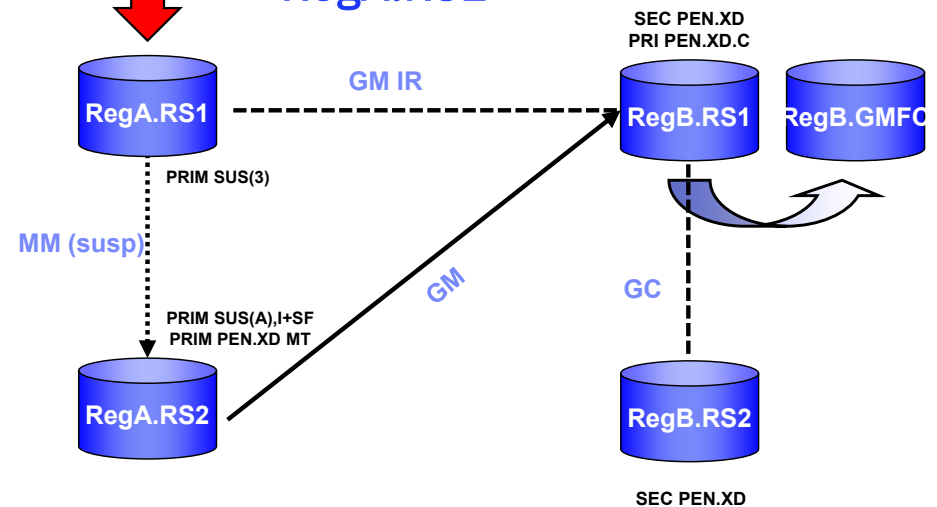
① After START SECONDARY
RegA.RS2 > RegA.RS1



③ After START SECONDARY
RegA.RS1 > RegA.RS2



② After HyperSwap RegA.RS2 > RegA.RS1



Unplanned primary disk outage – return to normal

- to initial configuration MGM cascaded

- Additional considerations to be covered post-lab

- if Kg-Sys moved to G0C2, move back to Site1 Controlling system as the 1st step of the return to normal procedure

- Used scripts:

- Control script for START SECONDARY RegA.RS2 > RegA.RS1
 - Control script for Planned Hyperswap RegA.RS2 > RegA.RS1
 - Control script for START SECONDARY RegA.RS1 > RegA.RS2

Unplanned primary disk outage – return to normal

- to initial configuration MGM cascaded

- Option (2)

- Run control script on Kp-Sys to establish Metro Mirror RegA.RS2 > RegA.RS1 with START SECONDARY, entering a MT configuration
- Run control script on Kg-Sys with MGM4SITE_RESTORE procedure to move to a cascaded configuration with primary MM disks in RegA.RS2 and primary GM disks in RegA.RS1
- Run control script in Kg-Sys with MGM4SITE_SWAP procedure to move primary MM disks back to RegA.RS1 and primary GM disks to RegA.RS2 maintaining an MGM 4-site cascaded configuration



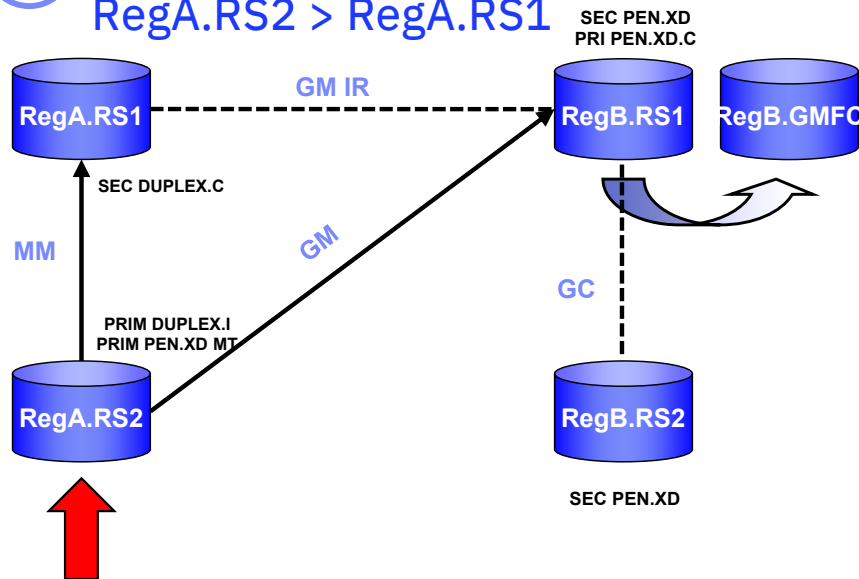
Maintains HA and DR readiness as much as possible



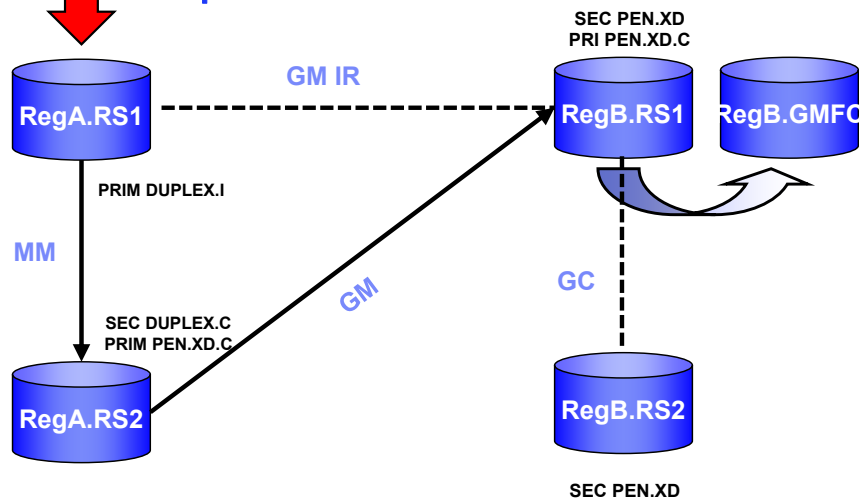
More reconfiguration steps

Return to normal – MGM cascaded - Option (2)

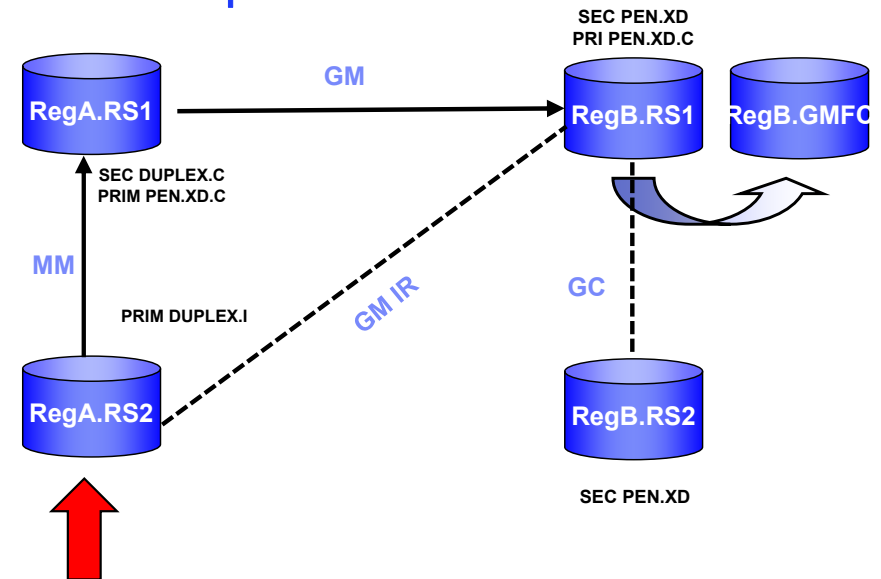
① After START SECONDARY
RegA.RS2 > RegA.RS1



③ After MGM4SITE_SWAP
procedure



② After MGM4SITE_RESTORE
procedure

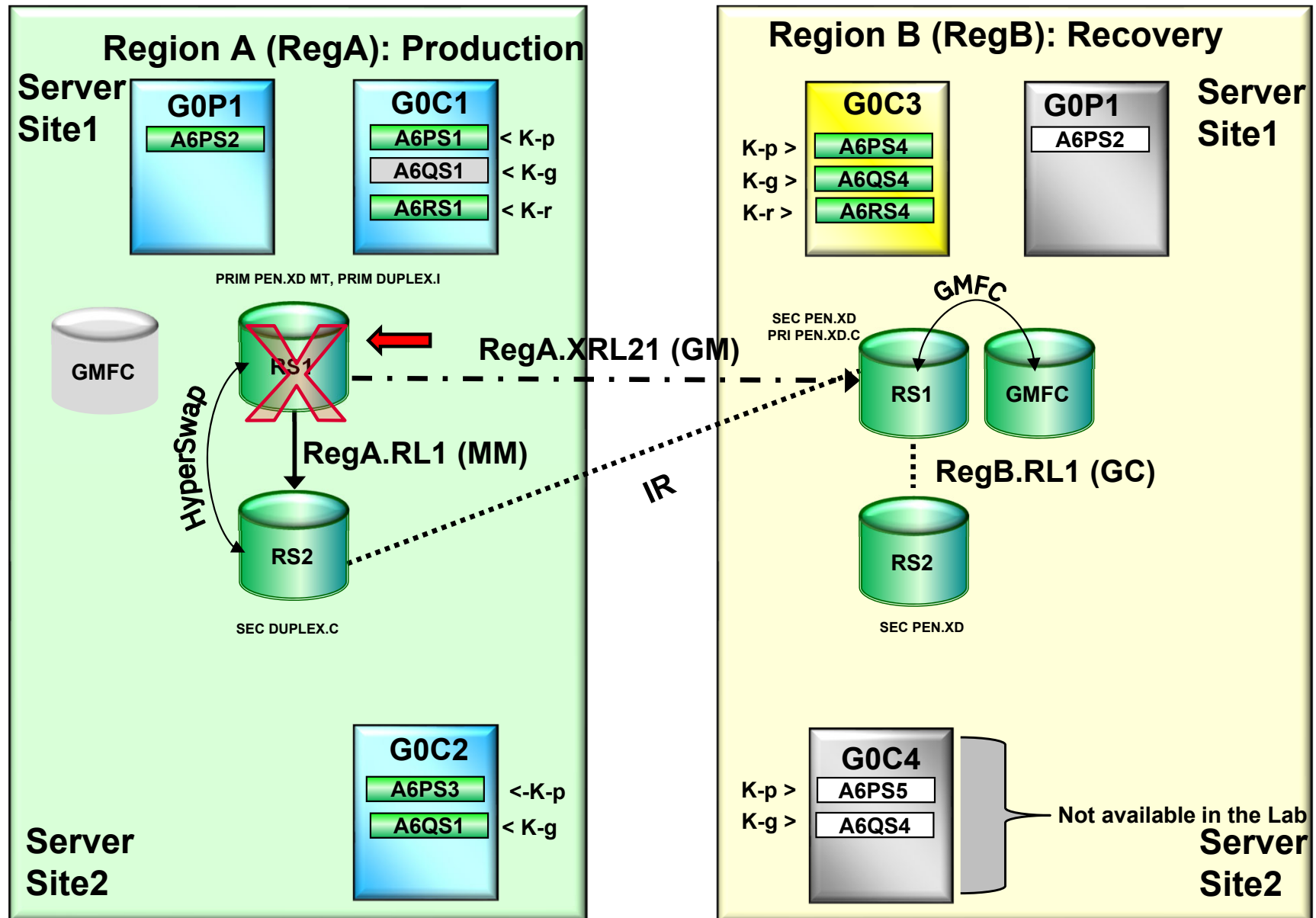


Unplanned primary disk outage

- triggering a HyperSwap with initial configuration MGM cascaded
- Additional considerations to be covered post-lab
 - if Kg-Sys moved to G0C2, move back to Site1 Controlling system as the 1st step of the return to normal procedure
- Used scripts:
 - Takeover script SWAPSITE1
 - Control script for START SECONDARY RegA.RS2 > RegA.RS1
 - Control script for MGM4SITE_RESTORE procedure
 - Control script for MGM4SITE_SWAP procedure

Unplanned primary disk outage

- triggering a HyperSwap with initial configuration MGM multi-target
 - Starting configuration



Unplanned primary disk outage

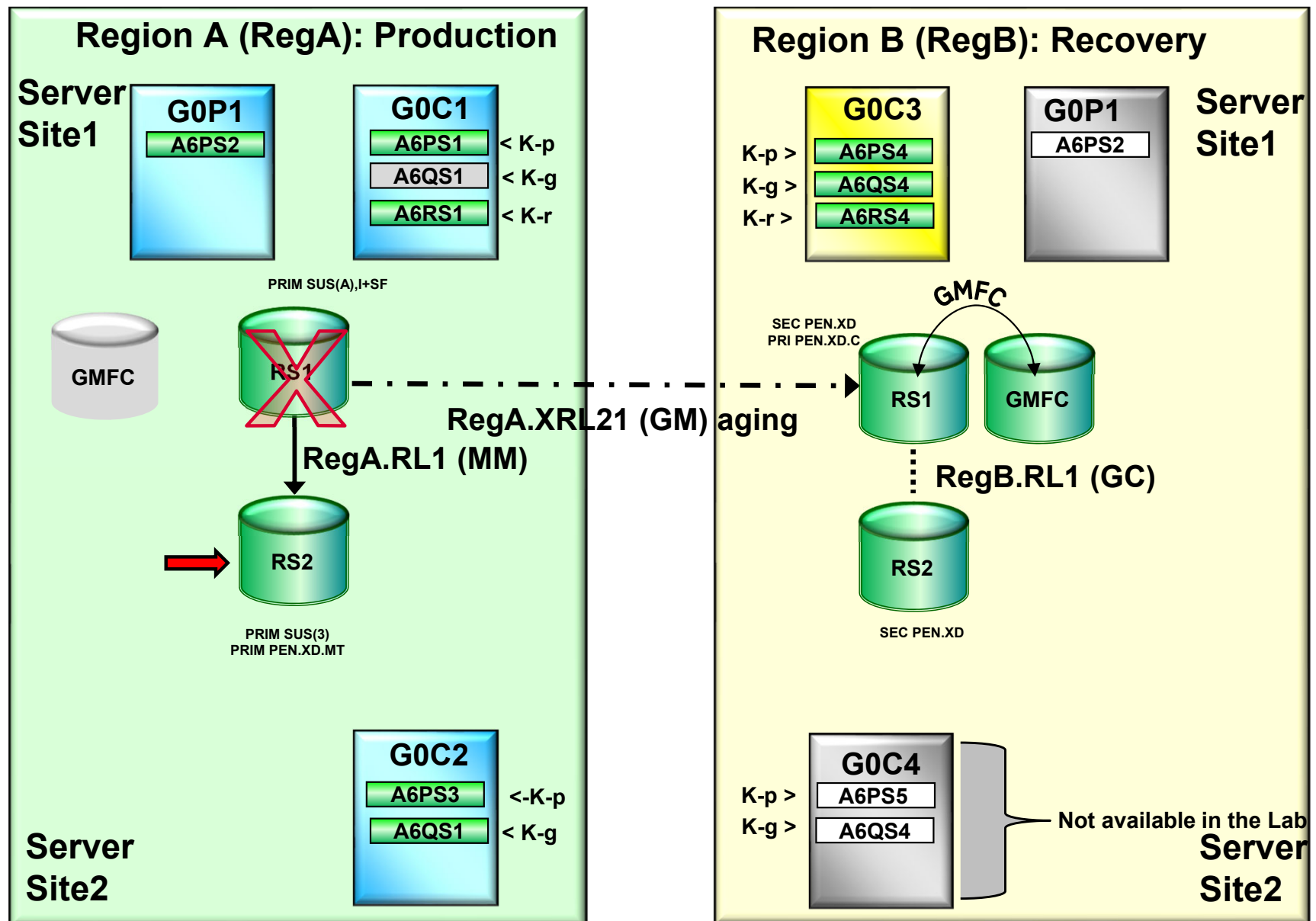
- triggering a HyperSwap with initial configuration MGM multi-target

- High level steps

- Primary disk failure (SETIOS HYPERSWAP) triggered unplanned HyperSwap from RS1 to RS2 in region RegA
- Metro Mirror RegA.RS1 > RegA.RS2 is frozen (FRZ)
- Application system GOP1 running on RegA.RS2
- DR position is aging, since GM session is running from failed RegA.RS1 disks
- Run control script on Kg-Sys with MGM4SITE_MOVEGM procedure to move GM to run from RegA.RS2 disks to regain DR

Unplanned Primary Disk Outage

- Triggering a Hyperswap with initial configuration MGM multi-target
- Intermediate configuration after HyperSwap



Unplanned primary disk outage – return to normal

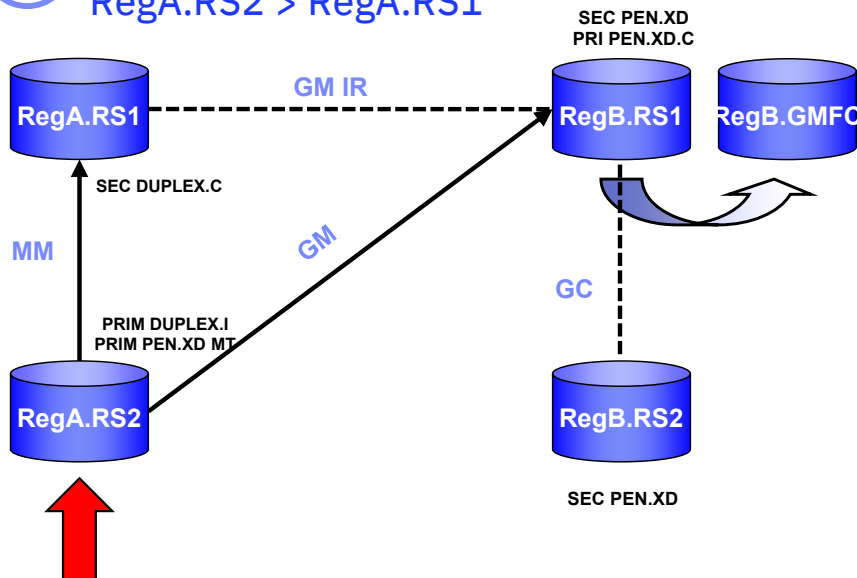
- to initial configuration MGM multi-target

- How to return to normal

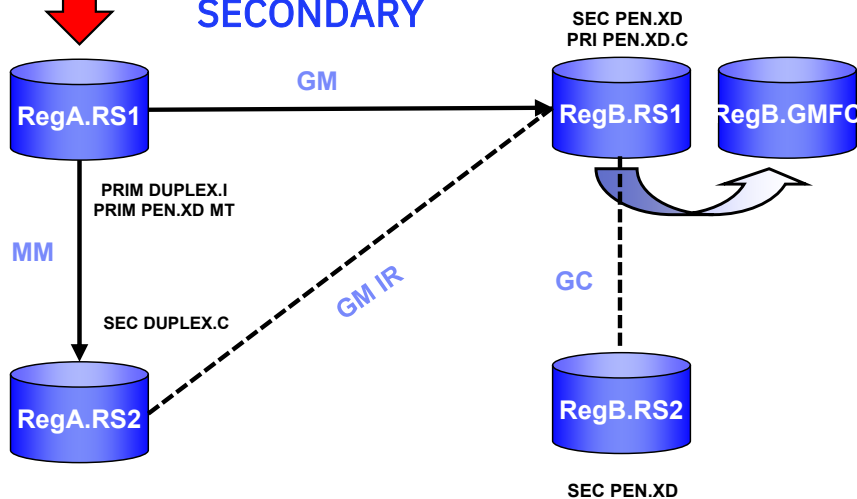
- Establish Metro Mirror RegA.RS2 > RegA.RS1 with START SECONDARY, entering a MT configuration
- Run MGM4SITE_RESTORE procedure to move to a cascaded configuration with primary MM disks in RegA.RS2 and primary GM disks in RegA.RS1
- Run HyperSwap RegA.RS2 > RegA.RS1 and establish Metro Mirror between RegA.RS1 > RegA.RS2 with START SECONDARY

Return to normal – MGM multi-target

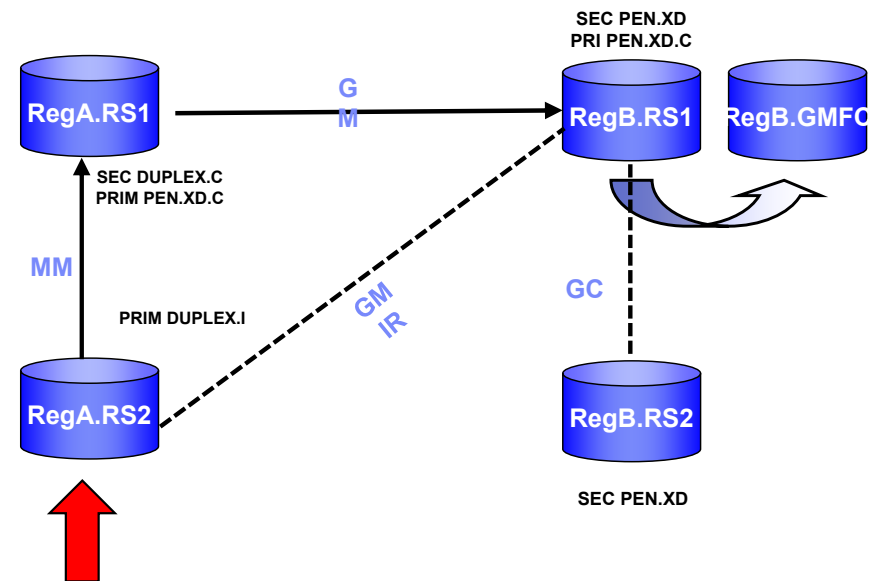
① After START SECONDARY
RegA.RS2 > RegA.RS1



③ After Hyperswap RegA.RS2 > RegA.RS1 and START SECONDARY



② After MGM4SITE_RESTORE procedure



Unplanned primary disk outage

- triggering a HyperSwap with initial configuration MGM multi-target

- Additional considerations to be covered post-lab

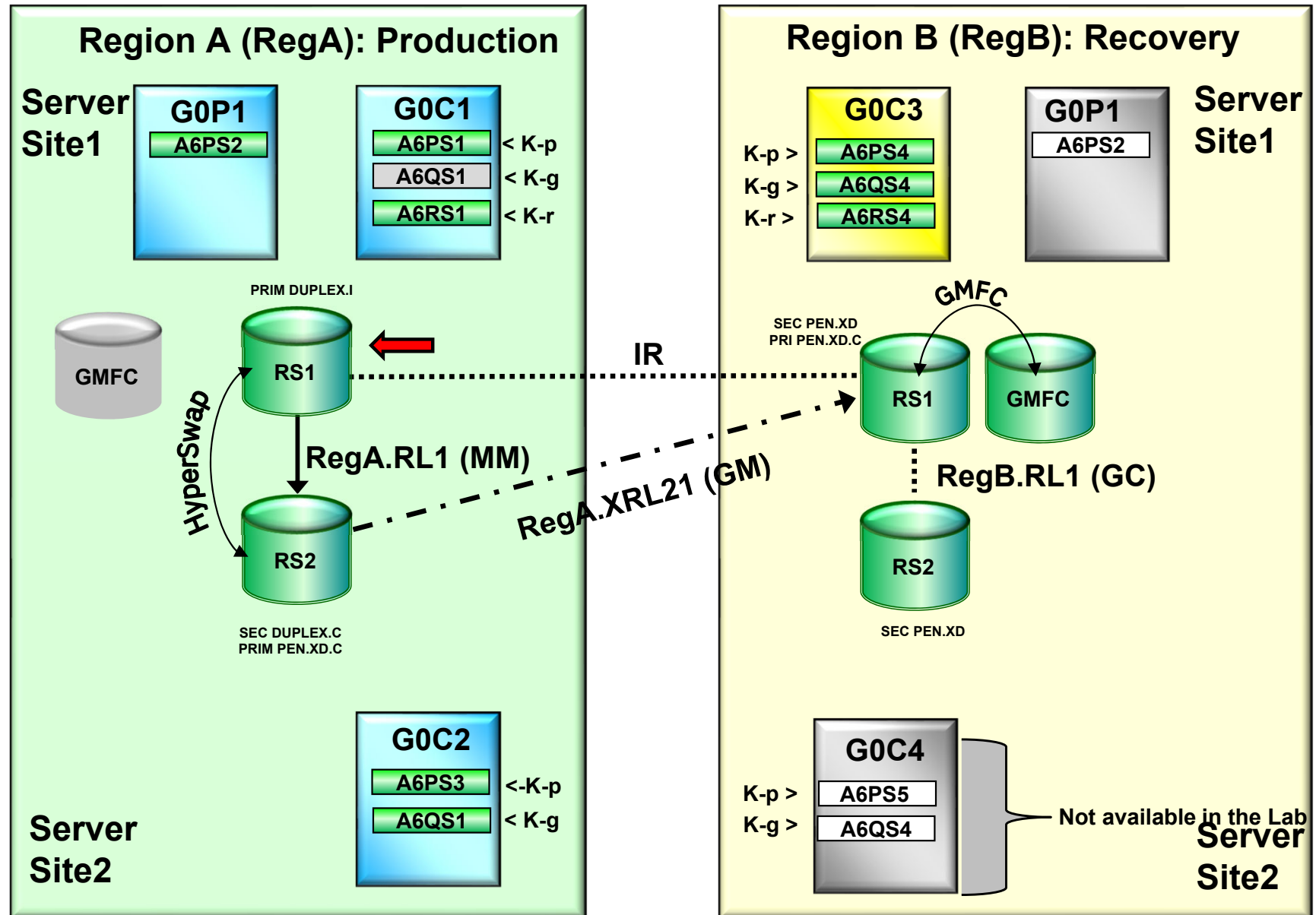
- in a multi-target configuration Kg-Sys runs on RS2 disks
 - move back of Kg-Sys not necessary

- Used scripts:

- Takeover script SWAPSITE1
 - Control script for START SECONDARY RegA.RS2 > RegA.RS1
 - Control script for MGM4SITE_RESTORE procedure
 - Control script for Planned Hyperswap RegA.RS2 > RegA.RS1
 - Control script for START SECONDARY RegA.RS1 > RegA.RS2

Planned Primary disk maintenance

- Planned Hyperswap with initial configuration MGM cascaded
- Starting configuration



Planned primary disk maintenance

- planned HyperSwap with initial configuration MGM cascaded

- High level steps

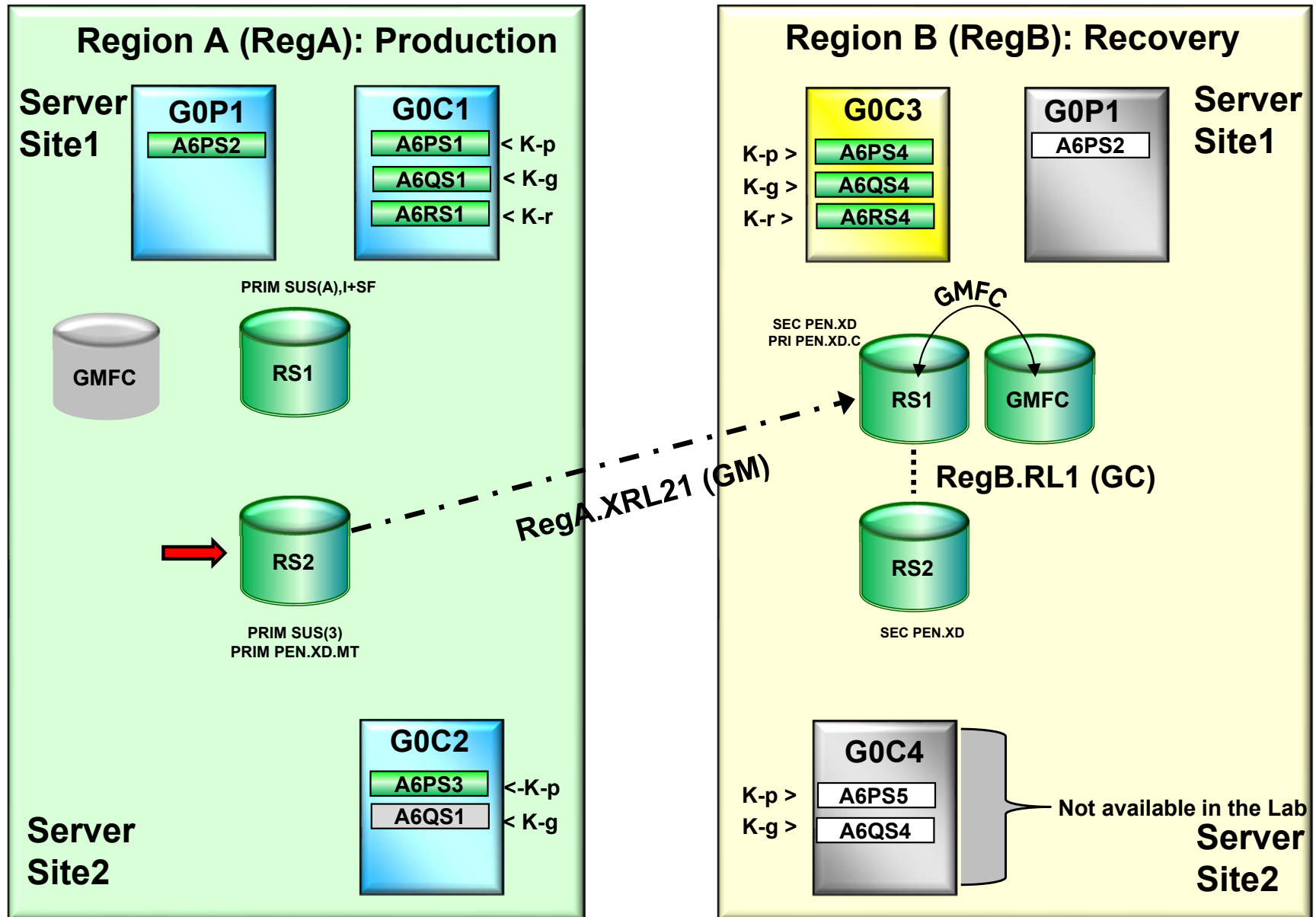
- Run Control script with planned HyperSwap RegA.RS1 > RegA.RS2
- Metro Mirror RegA.RS1 > RegA.RS2 is frozen (FRZ)
- Application system GOP1 running on RegA.RS2
- GM session RegA.RS2 > RegB.RS1 not moved, DR protection maintained

- Additional considerations to be covered pre-lab

- if SIMPLEX DASD in RS1 affected by maintenance, move Kg-Sys to G0C2

Planned Primary disk maintenance

- Planned Hyperswap with initial configuration MGM cascaded
- Starting configuration



Planned primary disk maintenance – return to normal

- to initial configuration MGM cascaded

■ Option (1)

- Establish Metro Mirror RegA.RS2 > RegA.RS1 with START SECONDARY, entering a MT configuration
- Planned HyperSwap RegA.RS2 > RegA.RS1, coming back to a cascaded configuration with primary MM disk in RegA.RS1
- Establish Metro Mirror RegA.RS1 > RegA.RS2 with START SECONDARY



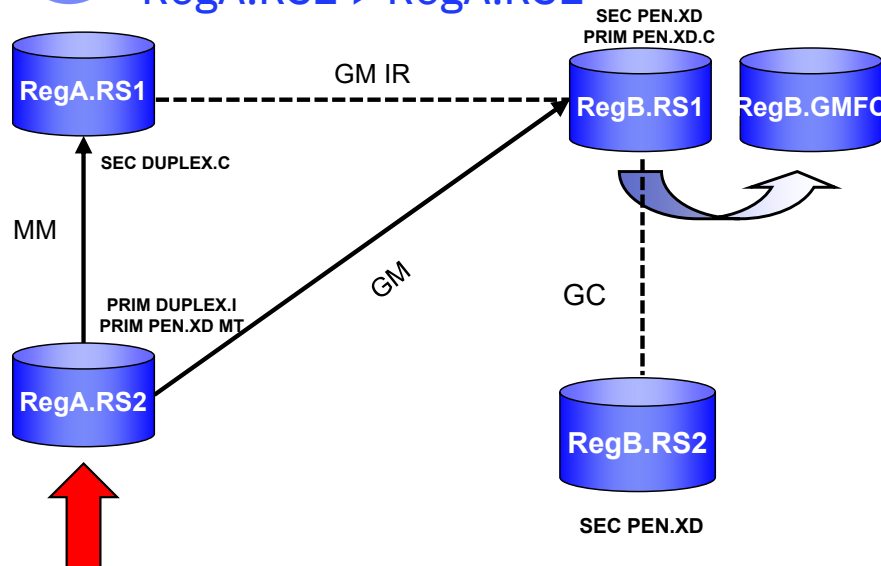
minimal management effort (shortest route)



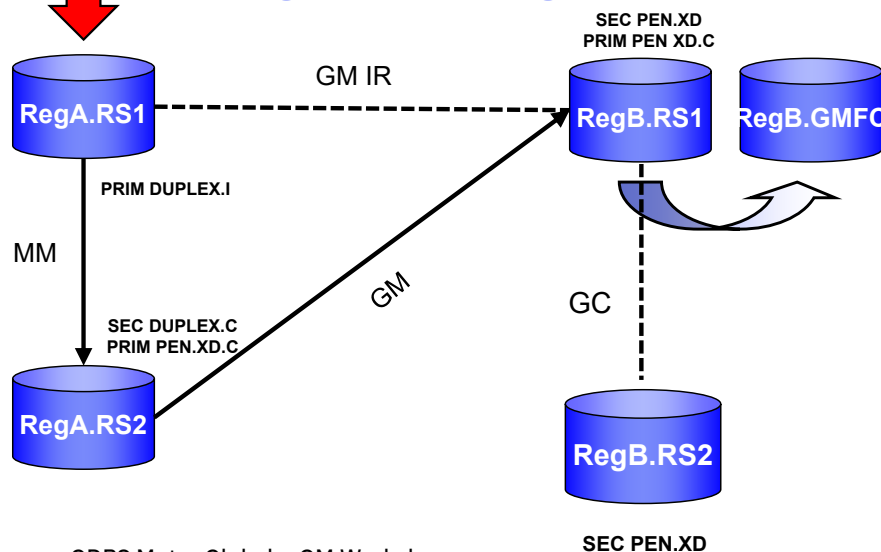
DR position ages from HyperSwap until START SECONDARY completes

Return to normal – MGM cascaded - Option (1)

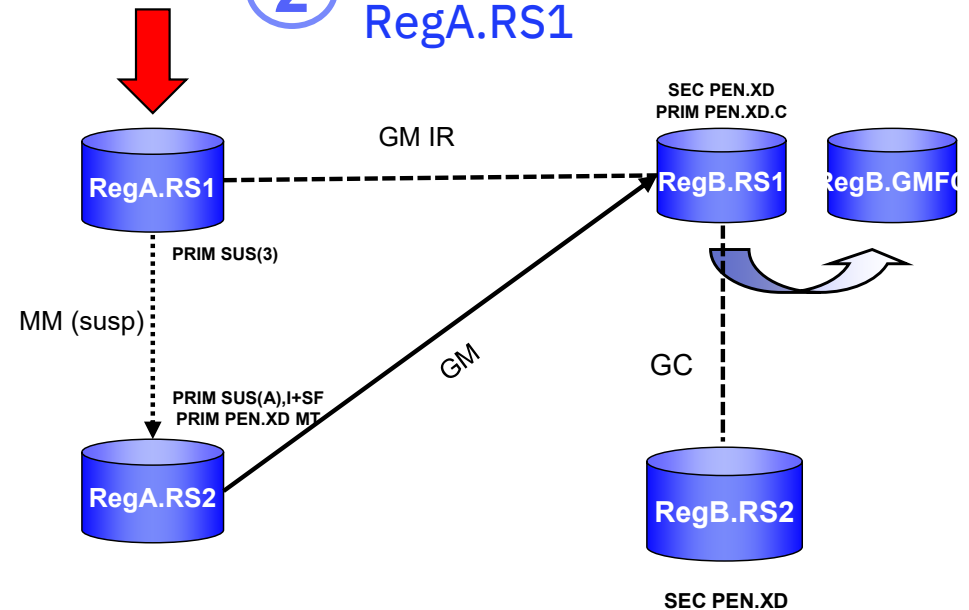
① After START SECONDARY
RegA.RS2 > RegA.RS1



③ After START SECONDARY
RegA.RS1 > RegA.RS2



② After HyperSwap RegA.RS2 > RegA.RS1



Planned primary disk maintenance

- planned HyperSwap with initial configuration MGM cascaded
- Additional considerations to be covered post-lab
 - if Kg-Sys moved to G0C2, move back to Site1 Controlling system as the 1st step of the return to normal procedure
- Used scripts:
 - Control script for Planned Hyperswap RegA.RS1 > RegA.RS2
 - Control script for START SECONDARY RegA.RS2 > RegA.RS1
 - Control script for Planned Hyperswap RegA.RS2 > RegA.RS1
 - Control script for START SECONDARY RegA.RS1 > RegA.RS2

Planned primary disk maintenance – return to normal

- to initial configuration MGM cascaded

▪ Option (2)

- Establish Metro Mirror RegA.RS2 > RegA.RS1 with START SECONDARY, entering a MT configuration
- Run MGM4SITE_RESTORE procedure to move to a cascaded configuration with primary MM disks in RegA.RS2 and primary GM disks in RegA.RS1
- Run MGM4SITE_SWAP procedure to move primary MM disks back to RegA.RS1 and primary GM disks to RegA.RS2 maintaining an MGM 4-site cascaded configuration



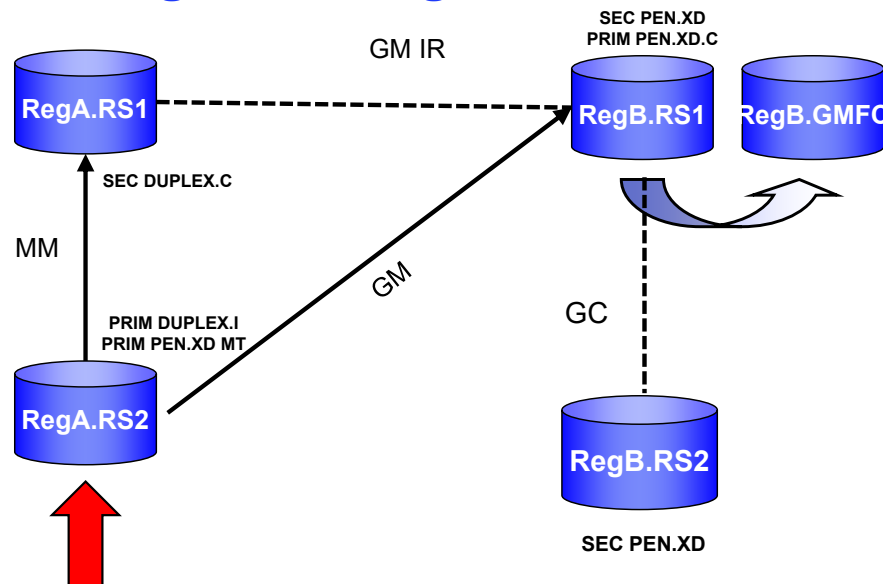
Maintains HA and DR readiness as much as possible



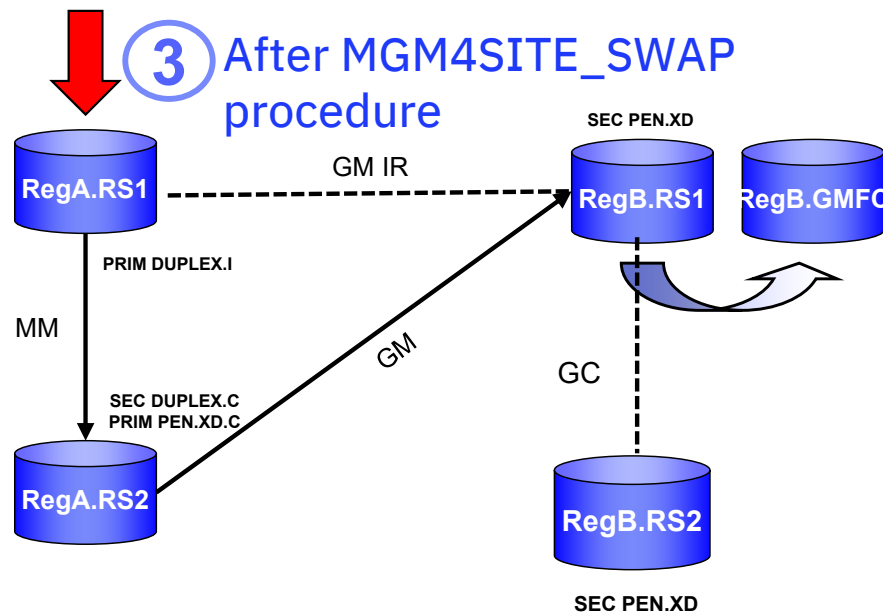
More reconfiguration steps

Return to normal – MGM cascaded - Option (2)

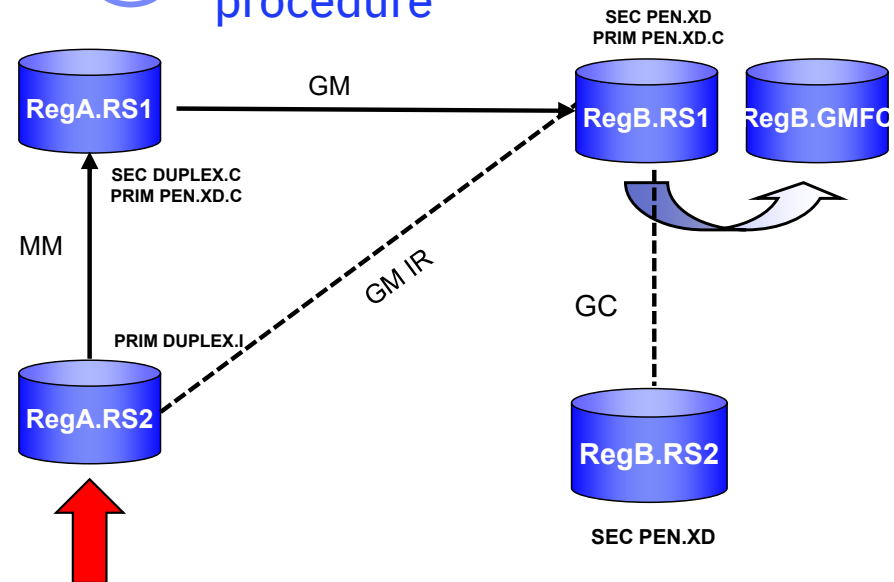
① After START SECONDARY RegA.RS2 > RegA.RS1



③ After MGM4SITE_SWAP procedure



② After MGM4SITE_RESTORE procedure

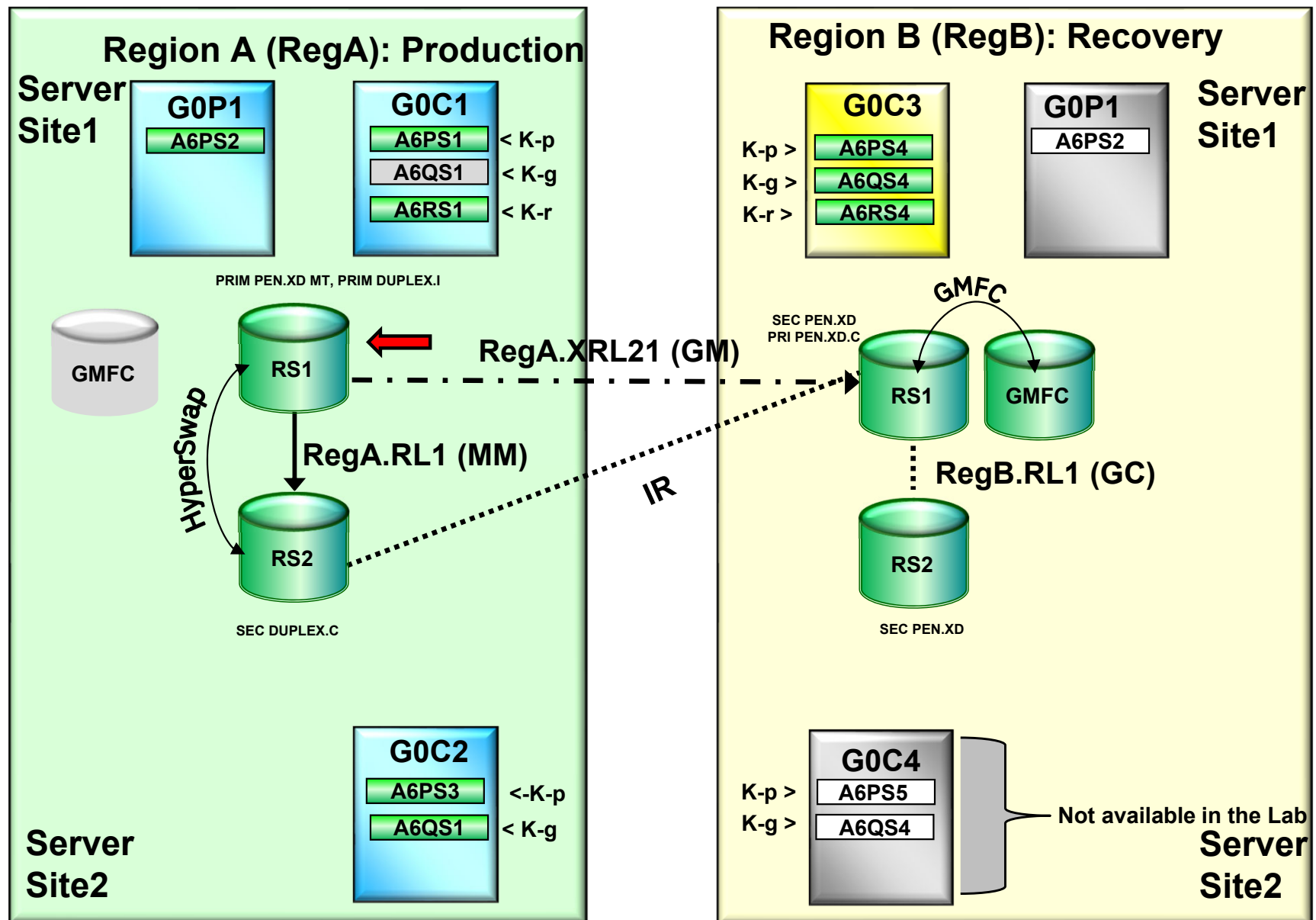


Planned primary disk maintenance

- planned HyperSwap with initial configuration MGM cascaded
- Additional considerations to be covered post-lab
 - if Kg-Sys moved to G0C2, move back to Site1 Controlling system as the 1st step of the return to normal procedure
- Used scripts:
 - Control script for Planned Hyperswap RegA.RS1 > RegA.RS2
 - Control script for START SECONDARY RegA.RS2 > RegA.RS1
 - Control script for MGM4SITE_RESTORE procedure
 - Control script for MGM4SITE_SWAP procedure

Planned primary disk outage

- Planned HyperSwap with initial configuration MGM multi-target
 - Starting configuration



Planned primary disk maintenance

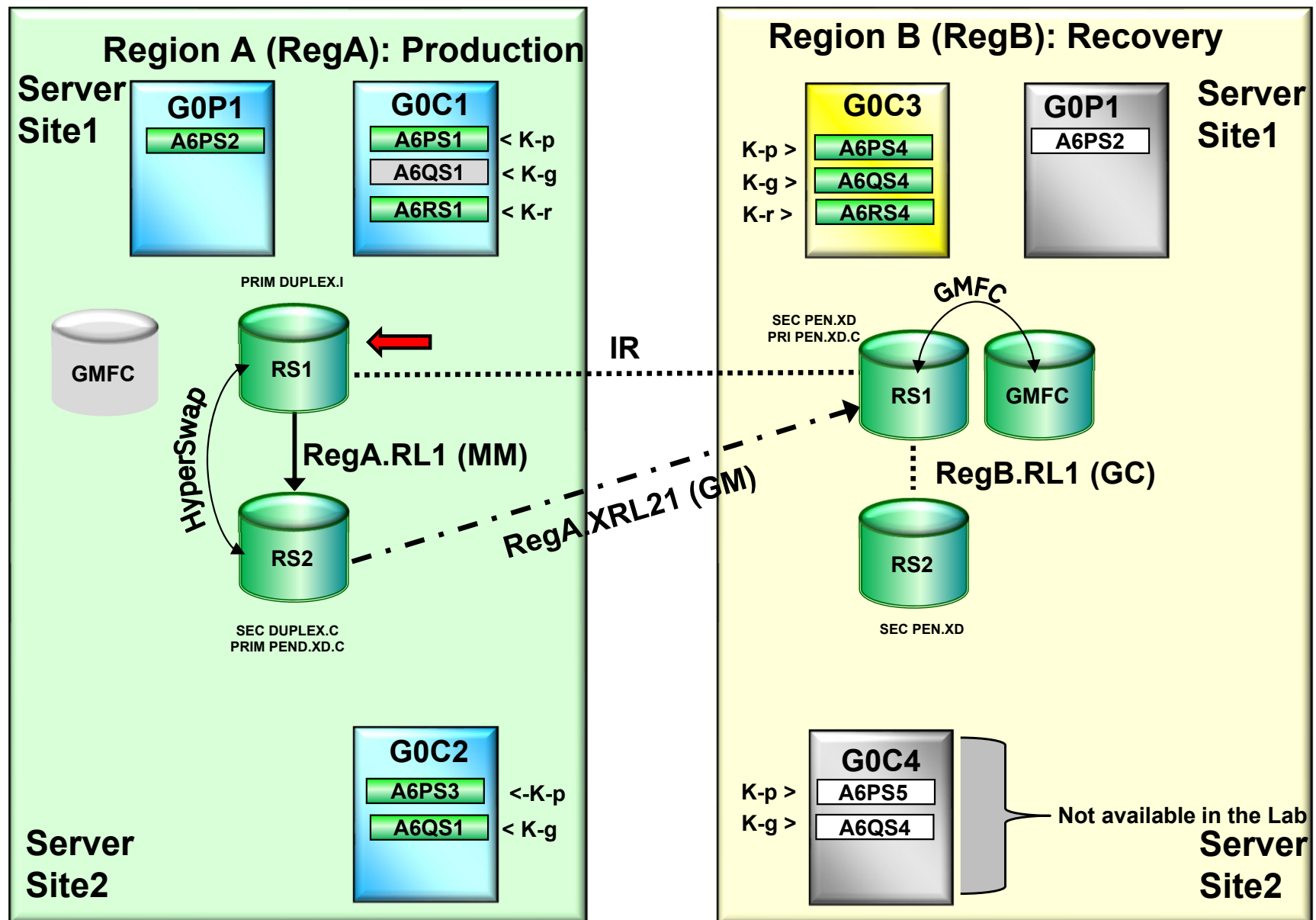
- planned HyperSwap with initial configuration MGM multi-target

- High level steps

- Run MGM4SITE_RESTORE procedure to avoid aging DR after HyperSwap, entering a cascading configuration
- if SIMPLEX DASD in RS1 affected by maintenance, move Kg-Sys to G0C2
- Run Control script with planned HyperSwap RegA.RS1 > RegA.RS2
- Application system GOP1 running on RegA.RS2

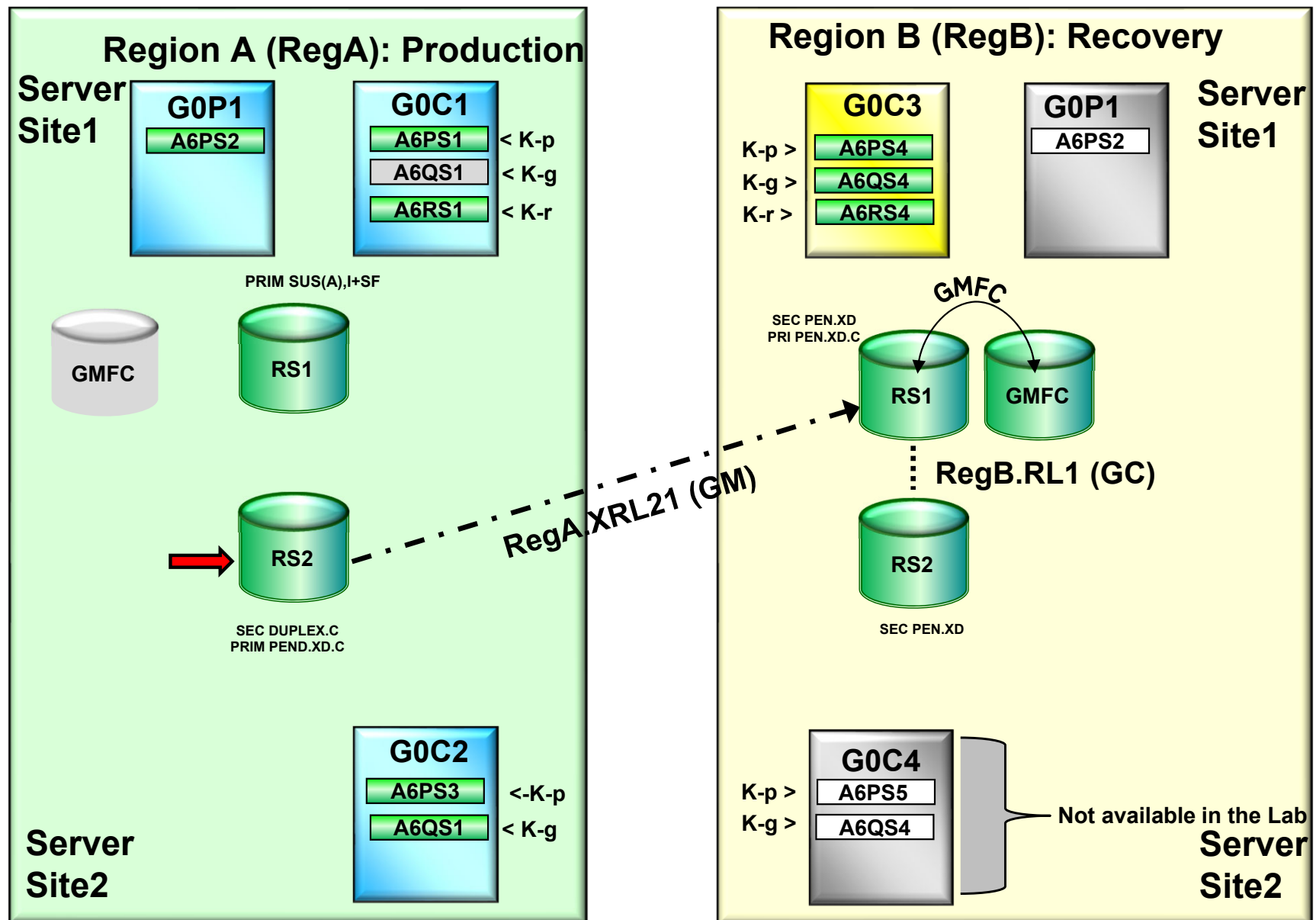
Planned primary disk outage

- Planned HyperSwap with initial configuration MGM multi-target
 - Intermediate configuration after MGM4SITE_RESTORE



Planned primary disk outage

- Planned HyperSwap with initial configuration MGM multi-target
 - Target Configuration



Planned primary disk maintenance – return to normal

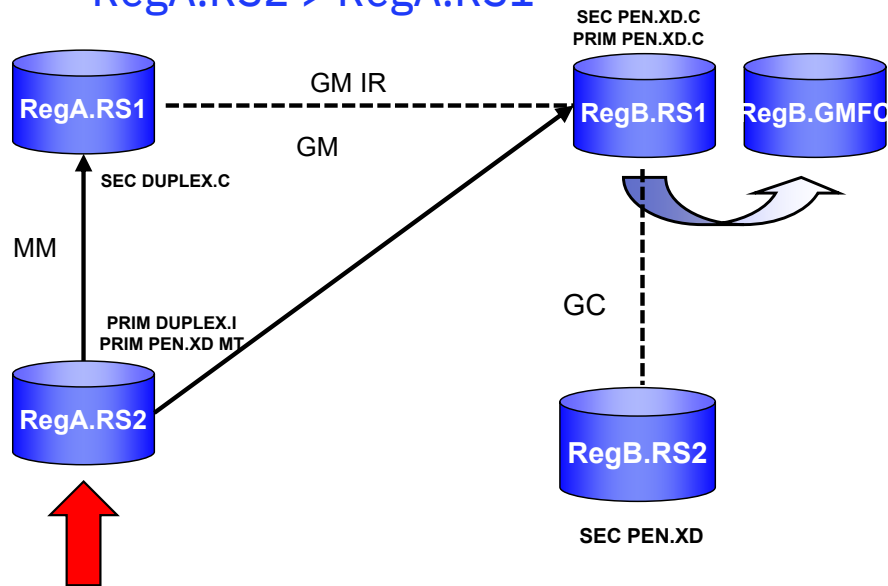
- to initial configuration MGM multi-target

- How to return to normal

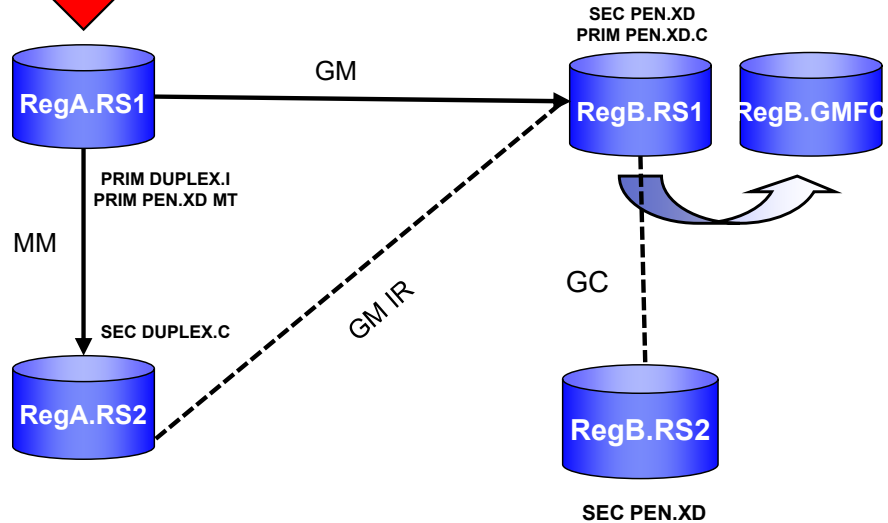
- Establish Metro Mirror RegA.RS2 > RegA.RS1 with START SECONDARY, entering a MT configuration
- Run MGM4SITE_RESTORE procedure to move to a cascaded configuration with primary MM disks in RegA.RS2 and primary GM disks in RegA.RS1
- Run HyperSwap RegA.RS2 > RegA.RS1 and establish Metro Mirror RegA.RS1 > RegA.RS2 with START SECONDARY

Return to normal – MGM multi-target

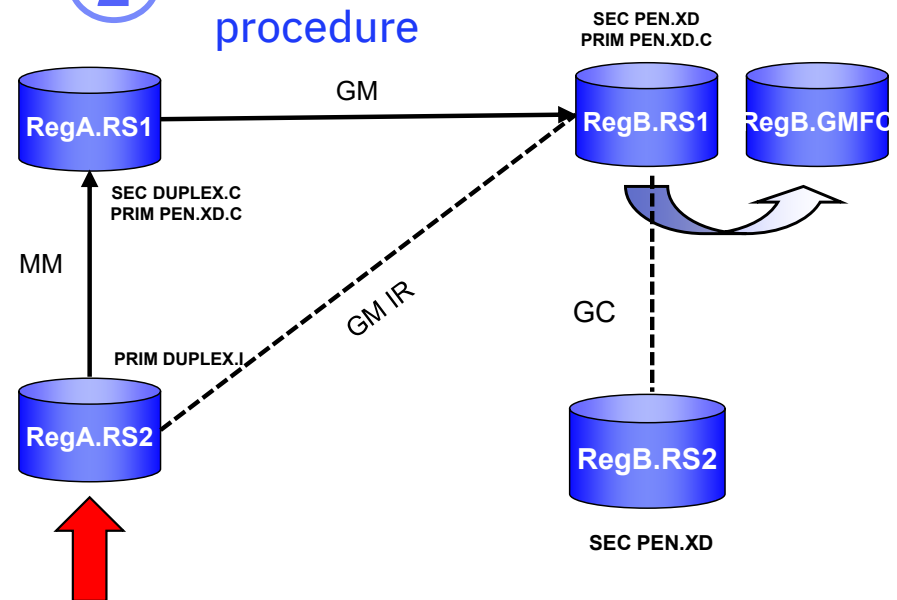
① After START SECONDARY
RegA.RS2 > RegA.RS1



③ After HYPERSWAP RegA.RS2 > RegA.RS1 and START SECONDARY



② After MGM3SITE_RESTORE procedure



Planned primary disk maintenance

- planned HyperSwap with initial configuration MGM multi-target
- Additional considerations to be covered post-lab
 - in a multi-target configuration Kg-Sys runs on RS2 disks
 - no move back of Kg-Sys necessary
- Used scripts:
 - Control script for MGM4SITE_RESTORE
 - Control script for Planned Hyperswap RegA.RS1 > RegA.RS2
 - Control script for START SECONDARY RegA.RS2 > RegA.RS1
 - Control script for MGM4SITE_RESTORE procedure
 - Control script for Planned Hyperswap RegA.RS2 > RegA.RS1
 - Control script for START SECONDARY RegA.RS1 > RegA.RS2

END OF LECTURE